Predictive Bayes with Large Language Models

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Background on Predictive Bayes

Exchangeability

Definition 1 (Exchangeable sequence)

A sequence of random variables $X_1, X_2, \ldots$ is infinitely exchangeable if for any $n \geq 1$ and a permutation $\sigma \in \mathbb{S}_n$,

$$(X_1, \ldots, X_n) \stackrel{d}{=} (X_{\sigma(1)}, \ldots, X_{\sigma(n)}). \tag{1}$$

de Finetti's representation theorem

Theorem 1 (de Finetti (Finetti, 1937; Hewitt et al., 1955))

Let $X_{1:\infty} := (X_1, X_2, \dots,)$ be a sequence of random variables taking values on a Polish space $\mathcal{X}$. $X_{1:\infty}$ is infinitely exchangeable if and only if there exists a random probability measure $\mathcal{P}$ defined on a space of measures $\mathfrak{M}(\mathcal{X})$ on $\mathcal{X}$ satisfying

$$\mathbb{P}(X_1 \in A_1, \dots, X_n \in A_n) = \int_{\mathfrak{M}(\mathcal{X})} \mu(A_1) \dots \mu(A_n) \mathcal{P}(\mathrm{d}\mu), \qquad (2)$$

for $n \geq 1$ and $A_1, \dots, A_n \in \mathcal{B}(\mathcal{X})$.

Exchangeability and de Finetti's theorem

In ML-friendly terms, exchangeability is equivalent to

$$\exists\,(F_\theta, \Pi_0) \text{ s.t. } \mathbb{P}(X_{1:n} \in A_{1:n}) = \int_\Theta \prod_{i=1}^{n} F_\theta(A_i)\Pi_0(\mathrm{d}\theta), \tag{3}$$

where (F_θ, Π_0) is a likelihood-prior pair. In other words,

- Exchangeability implies existence of a latent variable θ with prior π_0.
- Given θ, the data are i.i.d. from f_θ—θ explains all dependencies.

Bayesian models via predictive distributions

Instead of specifying a parameter with a prior and likelihood, given observed data, we instead elicit a sequence of predictive distributions,

$$P_n(A) := \mathbb{P}(X_{n+1} \in A | X_{1:n} = x_{1:n}) \text{ for } n \in \{0\} \cup \mathbb{N}, \quad A \in \mathcal{B}(\mathcal{X}), \tag{4}$$

with P_0 a probability measure on $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$. Under this model, the probability of observing a dataset $x_{1:n}$ is,

$$\mathbb{P}(X_{1:n} \in \mathrm{d}x_{1:n}) = \prod_{i=1}^{n} P_{i-1}(\mathrm{d}x_i). \tag{5}$$

Do the predictive distributions define a valid stochastic process $\{X_n\}_{n \in \mathbb{N}}$?

Bayesian models via predictive distributions

Theorem 2 (Ionescu-Tulcea)

Let P_0 and P_n be defined as in Eq. (4), where each P_n is measurable in $\sigma(X_{1:n})$. Then there exists a stochastic process $\{X_n\}_{n\in\mathbb{N}}$ such that

$$X_1 \sim P_0, \quad X_{n+1}\,|\,X_{1:n} \sim P_n, \tag{6}$$

and the law $\mathbb{P}$ of $\{X_n\}_{n\in\mathbb{N}}$ is uniquely determined.

Bayesian models via predictive distributions

- If $\{P_n\}$ were exchangeable, Theorem 1 guarantees the existence of (F_θ, Π_0).
- However, in practice, exchangeability is quite a strong requirement.
- Moreover, unless constructed to be exchangeable by design, it is hard to check whether $\{P_n\}$ is exchangeable.

Predictive resampling

Upon observing a context $X_{1:n} = x_{1:n}$, with a predictive sequence $\{P_n\}$,

- Draw $X_{n+1} \sim P_n$, append it to the context to get P_{n+1}.
- Draw $X_{n+2} \sim P_{n+1}$, append it to the context to get P_{n+2}.
- $\ldots$

The generated future data $X_{n+1}, X_{n+2}, \ldots$ induces a sequence of random predictive distributions $P_{n+1}, P_{n+2}, \ldots$.

Predictive resampling

What do we expect from this random distribution sequence?

Definition 2 (Predictive coherence (Fong et al., 2023))

Let $X_{1:n}$ be i.i.d. from some distribution F_0 and consider the predictives $\{P_{n+m}\}_{m\in\mathbb{N}}$ conditioned on $X_{1:n} = x_{1:n}$. The predictive coherence conditions are defined as follows.

- **Existence**: for each measurable A, the sequence $\{P_{n+m}(A)\}_{m\in\mathbb{N}}$ converges almost surely to $P_\infty(A)$ as $m \to \infty$, where P_∞ is a random distribution.
- **Unbiasedness**: $\mathbb{E}[P_\infty(A)\,|\,x_{1:n}] = P_n(A)$.

Predictive coherence

When is the coherence condition satisfied?

Definition 3 (Conditionally identically distributed (Berti et al., 2004))

The sequence $\{X_n\}_{n\in\mathbb{N}}$ is Conditionally Identically Distributed (C.I.D.) if for all $n, m \in \mathbb{N}$,

$$X_{n+m} \,|\, X_{1:n} \overset{\mathrm{d}}{=} X_{n+1} \,|\, X_{1:n}. \tag{7}$$

Equivalently, for each measurable A, the sequence $\{P_n(A)\}$ forms a martingale, that is, for all $n \in \mathbb{N}$,

$$\mathbb{E}[P_{n+1}(A) \,|\, X_{1:n}] = P_n(A) \quad \text{a.s.} \tag{8}$$

Predictive coherence

Theorem 3 (Berti et al. (2004))

Let F_N be a random empirical distribution constructed with the observed $x_{1:n}$ and random future $X_{n+1:N}$:

$$F_N := \frac{1}{N}\left(\sum_{i=1}^{n} \delta_{x_i} + \sum_{i=n+1}^{N} \delta_{X_i}\right). \tag{9}$$

Then $P_\infty = F_\infty$ almost surely.

Martingale posteriors

Let $\theta(F)$ be a functional that maps a distribution F into a quantity of interest. For instance, we may consider the generalized Bayes framework (Bissiri et al., 2016),

$$\theta(F) := \arg\min_{\theta} \int_{\mathcal{X}} \ell(\theta, x) F(\mathrm{d}x). \tag{10}$$

where $\ell(\theta, x)$ is a loss function depending on the quantity we wish to recover from P. Some examples:

- $\ell(\theta, x) = (\theta - x)^2$: $\theta(F)$ is the mean of F.
- $\ell(\theta, x) = |\theta - x|$: $\theta(F)$ is the median of F.
- $\ell(\theta, x) = -\log f_\theta(y)$ for some model density f_θ:

$$\theta(F) = \arg\min_{\theta} D_{\mathrm{KL}}[F \| F_\theta]. \tag{11}$$

Martingale posteriors

Definition 4 (Martingale posterior (Fong et al., 2023))

The martingale posterior is defined as,

$$\Pi_\infty(\theta_\infty \in A | x_{1:n}) = \int \mathbb{1}\{\theta(F_\infty) \in A\} \Pi(\mathrm{d}F_\infty | x_{1:n}). \tag{12}$$

As an approximation to the martingale posterior, the finite martingale posterior for $N \in \mathbb{N}$ is defined as,

$$\Pi_N(\theta_N \in A | x_{1:n}) = \int \mathbb{1}\{\theta(F_N) \in A\} \Pi(\mathrm{d}F_N | x_{1:n}). \tag{13}$$

The martingale posterior distribution

- The martingale posterior includes Bayes posterior as a special case, when $\{P_n\}$ is exchangeable.
- It generalizes the Bayesian posterior to a broader class of coherent predictive models, where epistemic uncertainty is represented through uncertainty about the unobserved future data.
- The target quantity is defined as a functional $\theta(F_\infty)$, for instance through a suitable loss function ℓ.

LLMs as Predictive Sequences

In-context learning

In-Context Learning (ICL) (Brown et al., 2020) is a remarkable property of Large Language Models (LLMs), where a pretrained LLM takes an observed dataset $\mathcal{D} := \{(x_i, y_i)\}_{i=1}^n$ as a context and predicts for a new query x_{n+1} without any retraining or fine-tuning.

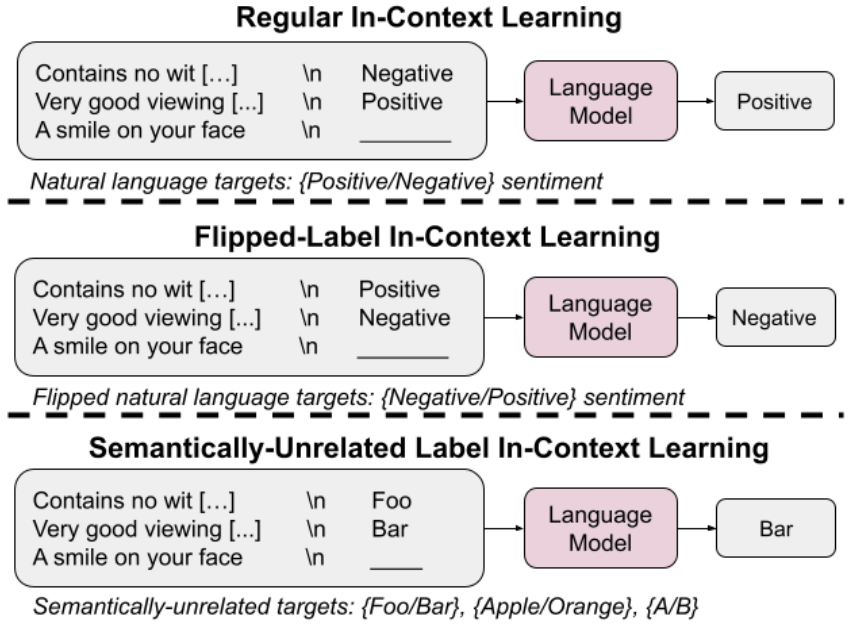


Figure 1: Source: Wei et al. (2023).

Autoregressive prediction in LLMs

A LLM can be viewed as a predictive sequence computed by a neural network.

$$\mathbb{P}(W_{n+1} = w_{n+1} \,|\, W_{1:n} = w_{1:n}) = \mathrm{softmax}(\mathtt{nn}(w_{1:n})). \tag{14}$$

There is no explicit "latent parameter" θ, for instance,

$$\mathbb{P}(W_{n+1} = w_{n+1} \,|\, W_{1:n} = w_{1:n}) = \int f_\theta(w_{n+1}) \pi_n(\theta \,|\, w_{1:n}) \mathrm{d}\theta. \tag{15}$$

But does it implicitly define θ, likelihood f_θ, and a prior π_0 under the hood?

ICL as an implicit Bayesian inference

- Several works conceptualize ICL as Bayesian inference over an implicitly defined latent parameter (concept) θ (e.g., Xie et al. (2022), Wang et al. (2023), Jiang (2023), Ye et al. (2024), and Falck et al. (2024)).
- Given an example dataset $Z_{1:n}$ where $Z_i := (X_i, Y_i)$ (input X_i and answer Y_i), an LLM performs ICL by sampling from the predictive distribution,

$$P_n(y_{n+1}) := \mathbb{P}(Y_{n+1} = y_{n+1} \,|\, X_{n+1} = x_{n+1}, Z_{1:n} = z_{1:n}), \tag{16}$$

where each Z_i can be a token sequence, and the examples $Z_{1:n}$ are assumed to be exchangeable.

ICL as an implicit Bayesian inference

- A Bayesian hypothesis for ICL states that there exists an underlying latent concept θ such that,

$$P_n(y_{n+1}) = \int f_\theta(y_{n+1})\pi_n(\theta \,|\, x_{n+1}, z_{1:n})\mathrm{d}\theta, \tag{17}$$

  with a proper likelihood f_θ and prior π_0.

- Why is this important? If this is true, the LLM prediction uncertainty isn't just aleatoric (due to data randomness), but is also epistemic (due to lack of model knowledge).

A martingale perspective on ICL

- As we have seen, an exchangeable predictive sequence implies the existence of an underlying Bayesian model.
- Therefore, if the predictive sequence $\{P_n\}_{n\in\mathbb{N}}$ is exchangeable, the Bayesian hypothesis holds.
- However, verifying exchangeability is often intractable, as it involves checking probabilities over exponentially many permutations.

A martingale perspective on ICL

- Falck et al. (2024) propose testing the martingale property or C.I.D. of the LLMs instead.
- The martingale property or C.I.D. is a necessary condition for a predictive sequence to correspond to a Bayesian model.
- If a LLM is not C.I.D., it cannot arise from any Bayesian model.
- Conversely, even if the sequence satisfies the martingale property, this alone does not guarantee that the model is Bayesian.

A martingale perspective on ICL

Proposition 1 (Falck et al. (2024))

A sequence $\{Z_{n+1:n+m}\}$ satisfies the martingale property if and only if the following holds: for all $n', k \in \mathbb{N}$ and integrable functions g, h:

$$\mathbb{E}[(g(Z_{n'+k}) - g(Z_{n'+1}))h(Z_{n+1:n'}) \,|\, Z_{1:n}] = 0. \tag{18}$$

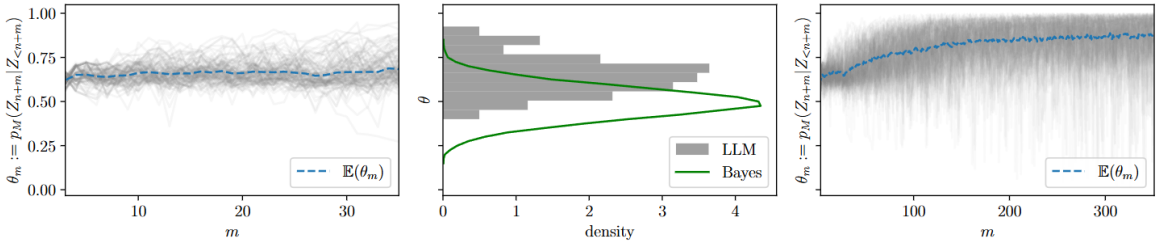
Corollary

If a sequence $\{Z_i\}_{i\in\mathbb{N}}$ satisfies the martingale property, then for all integers $n, n', k > 0$ and $n' > n$ it holds that:

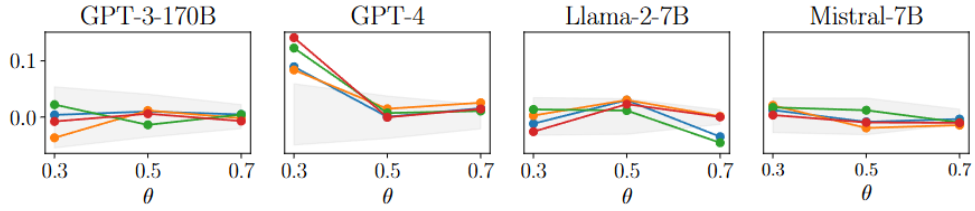
- $\mathbb{E}[g(Z_{n+1}) \,|\, Z_{1:n}] = \mathbb{E}[g(Z_{n+k} \,|\, Z_{1:n})]$ *for all integrable functions g,*
- $\mathbb{E}[(Z_{n'+k+1} - Z_{n'+1})Z_{n'}^\top \,|\, Z_{1:n}] = 0.$

A martingale perspective on ICL

LLMs generally fail the martingale test, suggesting that they are not Bayesian!



(b) $T_{2,k}$ for $k \in \{2, 3, 4, 5\}, n = 50, m = 25$



(d) $T_{2,k}$ for $k \in \{2, 3, 4, 5\}, n = 50, m = 100$

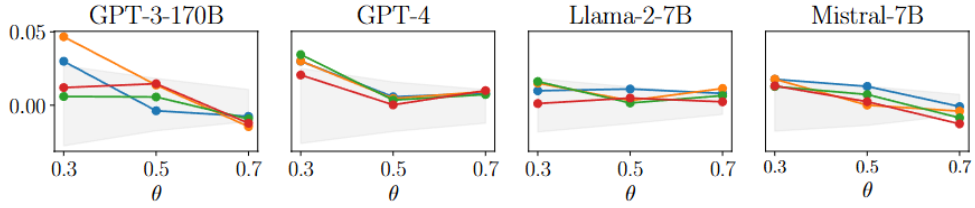


Figure 2: Source: Falck et al. (2024).

From Drift to Coherence: Stabilizing Beliefs in LLMs

From ICL to generic queries, from synthetic to real

Experiments in Falck et al. (2024):

We consider three types of synthetic datasets $z_{1:n}$:

- **Bernoulli**: $Z_i \sim \text{Bern}(\theta)$, where $\theta \in \{0.3, 0.5, 0.7\}$;
- **Gaussian**: $Z_i \sim \mathcal{N}(\theta, 1)$, where $\theta \in \{-1, 0, 1\}$;
- A synthetic **natural language** experiment representing a prototypical clinical diagnostic task, where $Z_i = (X_i, Y_i)$ indicate the presence or absence of a symptom and disease as a text string for the i-th patient, respectively. Further, $X_i \sim \text{Bern}(0.5), Y_i | X_i \sim \text{Bern}(0.3 + 0.4X_i)$.

A more realistic use case: asking a question Q to get an answer A!

From ICL to generic queries, from synthetic to real

- Turn a query Q into a prompt (a sequence of tokens) $\mathbf{x}_Q$.
- Let $p_\phi(\cdot|\mathbf{x}_Q)$ be a query-specific LLM prompted with $\mathbf{x}_Q$.
- How do we define predictive resampling? - let p_ϕ answer the same query repeatedly, while progressively conditioning on its own previous answers!

$$\begin{aligned} A_1 &\sim p_\phi(\cdot|\mathbf{x}_Q) := \theta_0(\mathbf{x}_Q), \\ A_2 &\sim p_\phi(\cdot|\mathbf{x}_Q, A_1) := \theta_1(\mathbf{x}_Q), \\ &\vdots \\ A_{n+1} &\sim p_\phi(\cdot|\mathbf{x}_Q, A_{1:n}) := \theta_n(\mathbf{x}_Q). \end{aligned} \tag{19}$$

- We call this a Prompted Predictive Resampling (PPR).

Prompted predictive resampling

An example prompt for multiple-choice QA tasks:

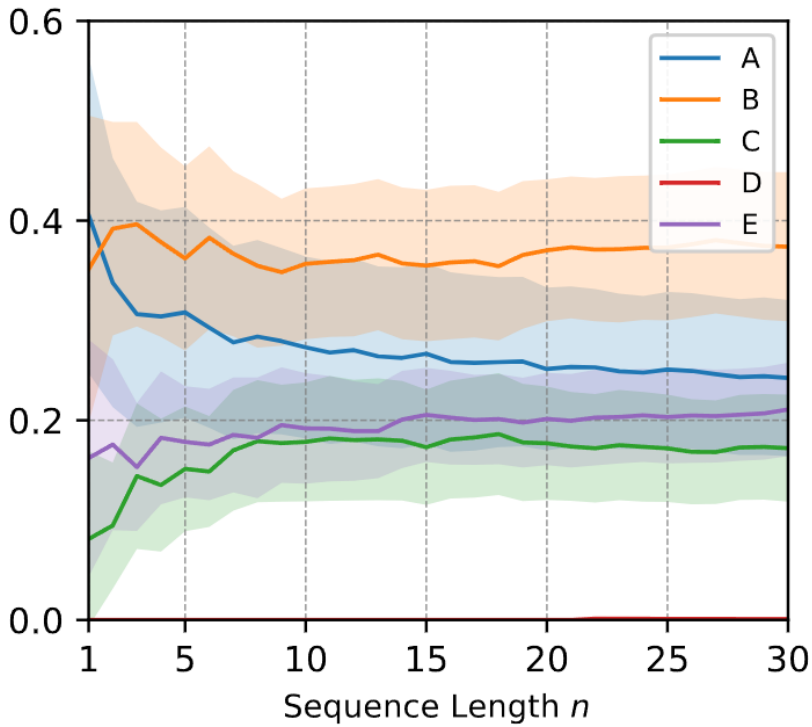
```
#### Generation examples
* If you think D is an answer for given question with 100% probability, then output might be :
D\nD\nD\nD\nD\n ...
* If you think B is the most plausible answer, but E can might also be an answer with small probability,
then output might be: B\nB\nB\nB\nE\nB\nB ...
* If you think either both A or C can be an answer with high probability, and the others(B, D, E, etc.) cannot
be the answer, then output might be : C\nA\nC\nC\nA\nA\n ... or A\nC\nC\nA\nA\nC\n ...

Now, generate the 100-line answer list for the given question below.
You must follow the output format and the generating rules!

Question:
'''
```

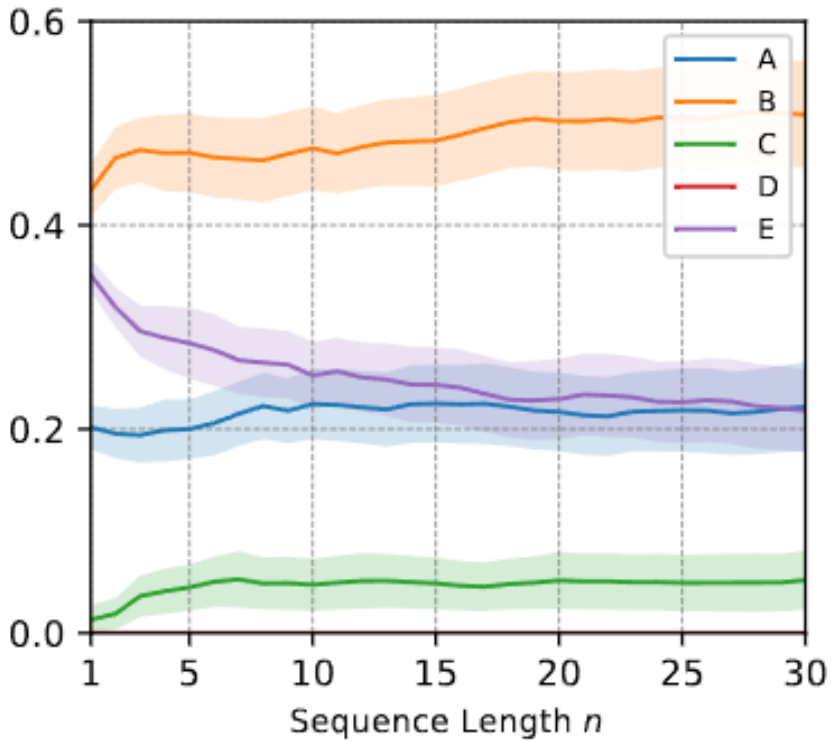
Prompted predictive resampling

The $\theta_n(\mathbf{x}_Q)$ trajectories obtained with Llama-3.1-8B (Grattafiori et al., 2024), for a question in CommonsenseQA benchmark (Talmor et al., 2019):



Prompted predictive resampling

"Seed answers": drawing $S_1, \ldots, S_n \overset{\text{i.i.d.}}{\sim} p_\phi(\cdot|Q)$ (without prompting) and appending $S_{1:n}$ to $\mathbf{x}_Q$ improves stabilization.



Prompted predictive resampling

Why does self-stabilization happen? Some intuitions:

- By conditioning on its own answer history $A_{1:n}$, the model obtains indirect evidence about the ambiguity of the question.
- Disagreement among early answers suggests that the question may be ambiguous or underspecified.
- If the early answers are consistent, the model tends to stabilize around that answer.
- Seed answers $S_{1:n}$ can accelerate this process by exposing early-stage uncertainty before self-resampling begins.

Prompted predictive resampling to martingale posteriors

From a sampled trajectory, we recover a target quantity $\theta(A_{1:N})$ as,

$$\bar{\theta}_N(A_{1:N}|\mathbf{x}_Q) = \underset{\theta\in\mathcal{P}(\mathcal{A})}{\arg\min} \int -\log \mathrm{Cat}(A|\theta) F_N(\mathrm{d}A|\mathbf{x}_Q), \tag{20}$$

which boils down to the empirical frequency vector computed from $A_{1:N}$. We repeat this over multiple trajectories and estimate the expectation.

$$A_{1:N}^{(1)}, \ldots, A_{1:N}^{(J)} \overset{\text{i.i.d.}}{\sim} p_\phi(\cdot|\mathbf{x}_Q), \quad \mathbb{E}[\theta_\infty|\mathbf{x}_Q] \approx \frac{1}{J}\sum_{j=1}^{J} \bar{\theta}_N(A_{1:N}^{(j)}|\mathbf{x}_Q). \tag{21}$$

Martingale amortization

We can finetune a pretrained LLM to reduce the early-step drifts.

Definition 5

Let $p^{(k)}_{\phi,n} := p_\phi(A_{n+k} \in \cdot \,|\, \mathbf{x}_Q, a_{1:n})$ denote k-step ahead marginal distribution. The self-consistency loss is defined as,

$$\mathcal{L}_{\text{SC}}(\phi) = \mathbb{E}_{k_1<k_2}\left[\mathbb{H}\left(\texttt{stopgrad}(\hat{p}^{(k_2)}_{\phi,n}, \hat{p}^{(k_1)}_{\phi,n}\right)\right], \tag{22}$$

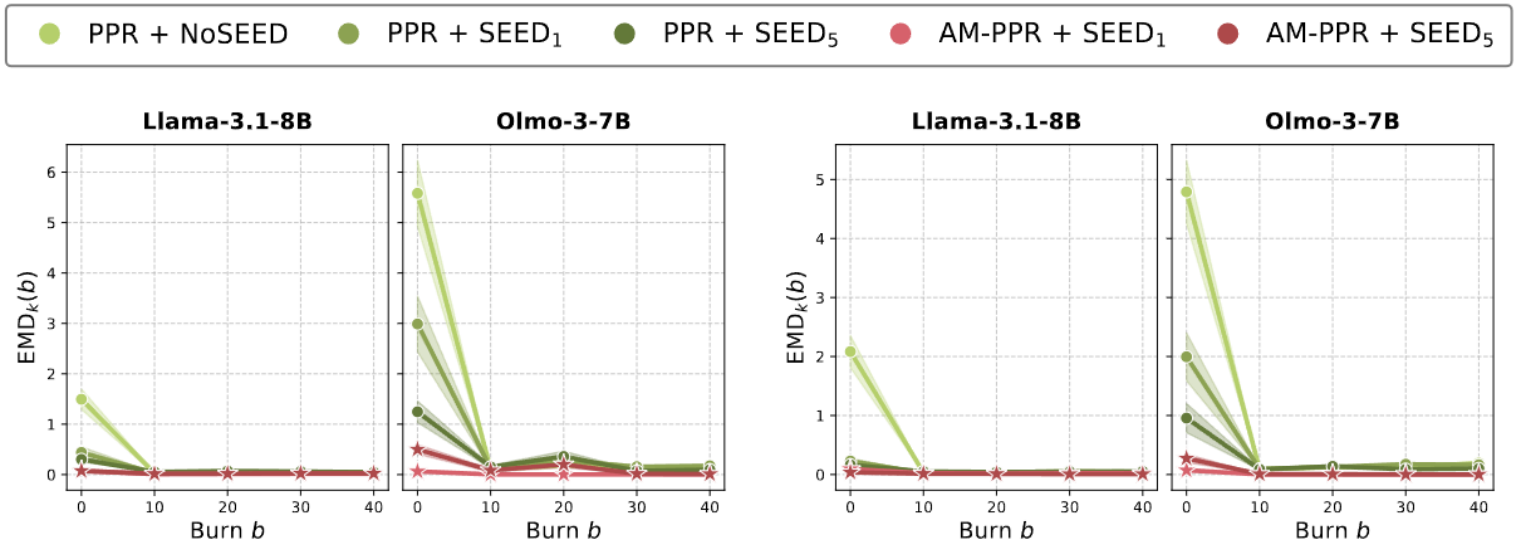
where $\hat{p}^{(k)}_{\phi,n}$ is the Monte-Carlo estimator of $p^{(k)}_{\phi,n}$,

$$\hat{p}^{(k)}_{\phi,n}(\cdot) = \frac{1}{J}\sum_{j=1}^{J} p_\phi(\cdot \,|\, A^{(j)}_{n+1:n+k-1}, a_{1:n}, \mathbf{x}_Q). \tag{23}$$

Experiments: measuring martingale violations

Measure the expected drift after the burn-in of b steps:

$$\mathrm{EMD}_k(b) := \mathbb{E}_{Q\sim\mathcal{D}}\left[\mathbb{E}_{A_{1:n}}\left[d\big(p^{(k)}_{\phi,n+b}, p^{(1)}_{\phi,n+b}\big)\right]\right] \tag{24}$$



(a) AI2-ARC-challenge

(b) TinyMMLU

Experiments: using martingale posteriors for QA tasks

Table 1. Accuracy and AUC score on TinyMMLU, AI2-ARC, CSQA inferred by LLAMA-3.1-8B

Method	TinyMMLU		AI2-ARC		CSQA	
	ACC	AUC	ACC	AUC	ACC	AUC
PPR + NoSEED	46.0	78.0	62.6	85.6	50.0	85.6
PPR + SEED_1	59.2	80.0	77.8	91.3	74.0	92.7
PPR + SEED_5	61.2	**83.1**	80.7	93.0	**78.0**	92.7
AM-PPR + SEED_1	**62.3**	**83.1**	**81.5**	92.9	**78.0**	**93.1**
AM-PPR + SEED_5	**62.2**	**85.1**	80.5	**94.2**	**77.0**	**94.2**
Direct-Query	61.2	81.6	80.5	**93.1**	**78.0**	91.9

Table 2. ECE, NLL and Brier Score on TinyMMLU, AI2-ARC, CSQA inferred by LLAMA-3.1-8B

Method	TinyMMLU			AI2-ARC			CSQA		
	ECE	NLL	BS	ECE	NLL	BS	ECE	NLL	BS
PPR + NoSEED	0.1763	1.2480	0.6247	0.1623	1.1199	0.5222	0.1273	1.5178	0.6430
PPR + SEED_1	0.1537	**1.0526**	0.5346	0.1695	**0.8936**	0.3968	0.1522	0.9109	0.3842
PPR + SEED_5	**0.1278**	**1.0721**	**0.5088**	0.1259	**0.7585**	0.3281	0.1967	**0.8832**	0.4324
AM-PPR + SEED_1	**0.1229**	2.2144	**0.5018**	0.1943	0.9240	0.3567	0.1139	0.9722	0.3537
AM-PPR + SEED_5	0.1568	1.2690	0.5123	**0.0845**	1.2752	**0.2872**	**0.0674**	**0.7886**	**0.3299**
Direct-Query	0.2457	5.4862	0.5631	**0.1251**	2.4481	**0.2978**	**0.1072**	3.0363	**0.3453**

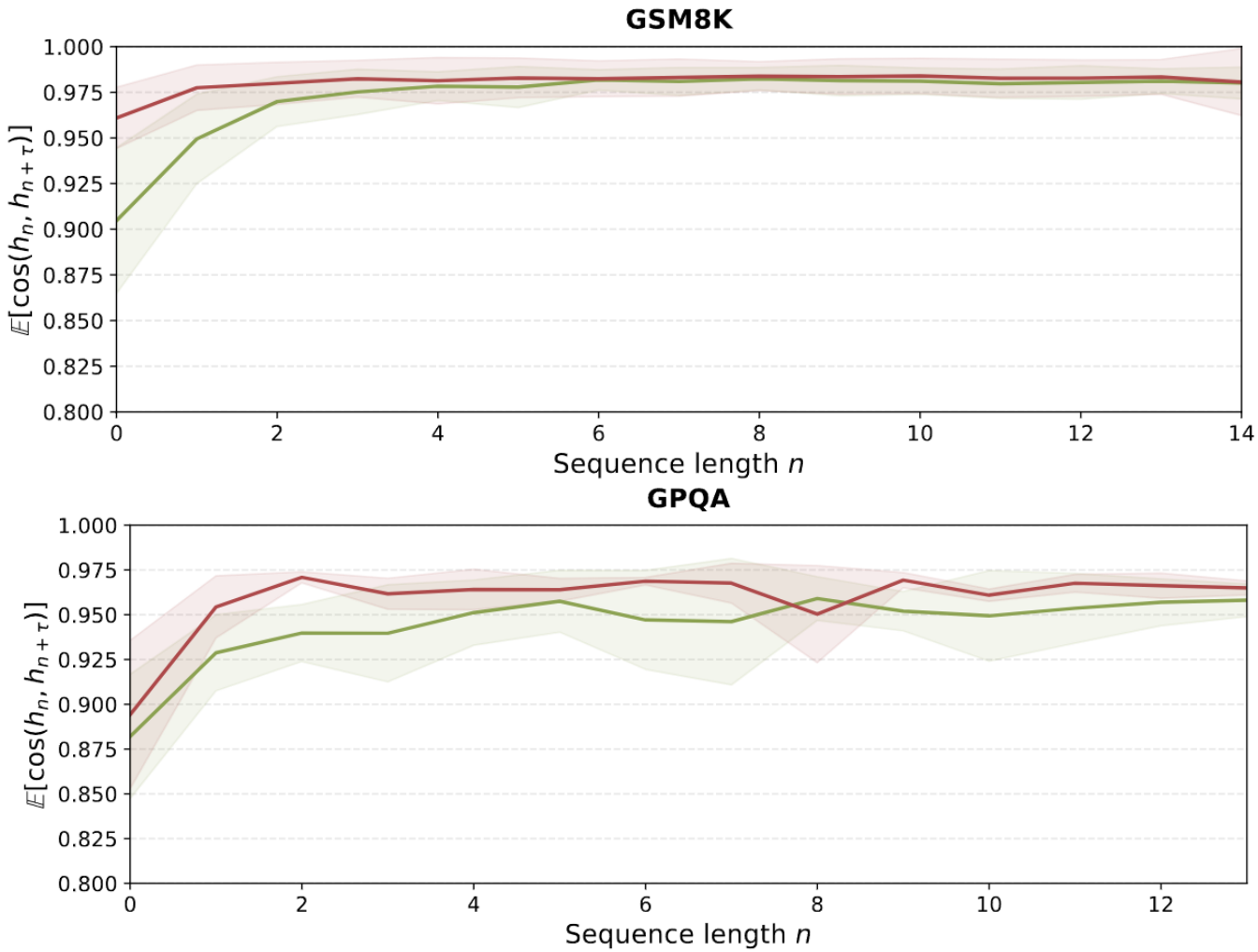
Extension to reasoning tasks

- For reasoning tasks, given a query Q, we let a model repeatedly generate a pair $R_n := (T_n, A_n)$ of CoT trace T_n and an open-ended answer A_n.
- We study the distribution of the embedding $H_n := \psi(R_n) \in \mathcal{H}$, or a random measure,

$$\mu_n(\mathbf{x}_Q) := p_\phi(H_{n+1} \in \cdot \,|\, H_{1:n}, \mathbf{x}_Q). \tag{25}$$

Extension to reasoning tasks

We observe a similar stabilization behavior.



Concluding remarks

Limitations:

- PPR relies on instruction-following ability; some models struggle to generate answers in the required format.
- The martingale condition diagnoses only a specific Bayesian predictive representation, not all forms of Bayesian uncertainty.
- Even with amortization, predictive resampling remains computationally expensive.

Concluding remarks

Future works:

- Study PPR in more diverse settings, such as multi-turn interactions and agentic workflows, across a broader class of models.
- Identify useful coherence conditions beyond the martingale property.
  - The martingale condition may be strong (cf. asymptotic C.I.D. (Battiston et al., 2025)).
  - It may also be too weak (e.g., deterministic predictive sequence).
- Design better post-hoc tuning methods for calibrating pre-trained models.

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